

Hosna Habibi Baghi

Machine Learning Engineer | NLP & Generative AI

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PROFESSIONAL SUMMARY

Machine Learning Engineer specializing in NLP and production AI systems. Experienced in building chatbot and contact-center pipelines with modern LLM frameworks (LangChain, LlamaIndex) and retrieval-augmented generation (RAG). Strong background in software engineering and model evaluation.

WORK EXPERIENCE

Software Formal Methods Researcher

Dec 2024 – Apr 2026

ER Research Center of Software and AI

Montreal, Canada

- Research on verification of temporal properties in timed event systems using formal methods.
- Built a simulation framework transforming UML state machines into Prolog representations for automated behavioral verification.
- Implemented a Python-Prolog engine to evaluate Linear Temporal Logic (LTL) properties over event traces.

Machine Learning Developer

Jul 2024 – Dec 2024

Prism

Tehran, Iran

- Designed and deployed NLP components for an intelligent contact-center platform using transformer models and LLMs.
- Built scalable conversation-analysis pipelines for intent detection, semantic search, and information extraction.
- Improved response relevance and reduced hallucinations through fine-tuning and retrieval-augmented generation (RAG).
- Shipped ML services to production via REST APIs; built evaluation frameworks combining automated metrics and human feedback.

Machine Learning Engineer

Sep 2023 – Jul 2024

FlowChat

Tehran, Iran

- Developed NLP models for production chatbot systems (BERT, RNNs, LSTMs) for intent classification and response generation.
- Designed end-to-end ML pipelines for training, evaluation, and deployment of conversational models.
- Improved chatbot performance iteratively using real-world interaction data and user feedback.

AI Researcher

Sep 2022 – Sep 2023

University of Guilan

Guilan, Iran

- Researched NLP and conversational AI using deep learning models (RNNs, LSTMs, BERT) for text classification, question answering, and response generation.
- Explored LLMs through prompt engineering and few-shot learning for multi-turn dialogue and topic analysis.

SKILLS

Programming: Python, Java, C++, JavaScript

Machine Learning & NLP: PyTorch, TensorFlow, Transformers, LLMs, Fine-tuning (LoRA, PEFT), RAG, Text Embeddings, Semantic Search, NER, Text Classification

LLM Frameworks & Tooling: LangChain, LangGraph, LlamaIndex, Vector Databases (FAISS, Pinecone), Agent-based Systems, Prompt Engineering

MLOps & Deployment: Docker, CI/CD, REST APIs, MLflow, Weights & Biases, Model Monitoring

LLM Evaluation: Offline/Online Evaluation, A/B Testing, Latency & Cost Optimization, Hallucination Mitigation

Tools & Other: Git, Linux/Unix, NumPy, Pandas, Matplotlib, React

EDUCATION

M.Sc., Computer Science

Apr 2026

Concordia University

Montreal, Canada

- Thesis: Verification of Temporal Properties in Timed Event Systems (Software Formal Methods).

B.Sc., Computer Engineering

Sep 2023

University of Guilan

Guilan, Iran

- Thesis: Blood Glucose Prediction Using Mixture of Experts.

PUBLICATIONS

Hosna Habibi Baghi, Ali Jannatpour, Constantinos Constantinides. "Verification of Linear Temporal Properties over Timed Events." International Conference on Software Engineering and Development (ICSED 2025), Springer CCIS, 2026. [\[Paper\]](#)

LANGUAGES

Persian (Native) • English (Fluent) • French (Beginner)